Design, Implementation, and Simulation of a Hybrid Wired–Wireless Network Using Cisco Packet Tracer

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1. Introduction

The current organizational settings require the network infrastructures that have the potential of sustaining a variety of communication needs, such as the high-speed completion of wired connections and the dependability of routing amidst various sub-networks, as well as the dynamic wireless services to the users on the move. The actual implementation in this report deals with the design of a hybrid wired and wireless IPv4 network that has three routed subnets. The project incorporates routers, switches, wireless infrastructure, servers, and end devices in a unified topology to facilitate flawless communication on all the network segments. In this report, the design methodology is described, and the IP address scheme based on the student identifier, the device configuration, the routing option, and the connectivity verification in all the subnets are elaborated.

This project aims to design and set up a hybrid network to include the components of wired and wireless networks in the project and also provide good inter-network communication. Its practical implementation is implemented with the Cisco Packet Tracer, which is a widely used simulation platform to learn concepts to do with network architecture, configuration, and routing behavior of various devices. The network has three logical subnets, including a wireless subnet, which is going to be used to offer mobile access, and two wired subnets, which support desktop clients and servers (Azizah et al., 2025). The topology will involve the use of three Cisco 1941 routers that, when combined, will form a routed backbone in order to provide efficient message forwarding between these subnets. RouterA is the core routing unit of the network where the wireless and wired segments are distributed to RouterA, RouterB, and RouterC, respectively.

2. IP Address Determination and Subnetting

2.1 Derivation of the Base Network Address

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Subnet	Purpose	Required Hosts	Allocated Mask	Network Address	Total Addresses	Usable Hosts	Broadcast Address
Subnet A	Wireless (AP + 3 Laptops)	4	**/29**	191.96.13.0	8	6	191.96.13.7
Subnet B	Wired (3 PCs + Server)	4	**/28**	191.96.13.8	16	14	191.96.13.23
Subnet C	Wired (4 PCs + Server)	5	**/27**	191.96.13.24	32	30	191.96.13.55

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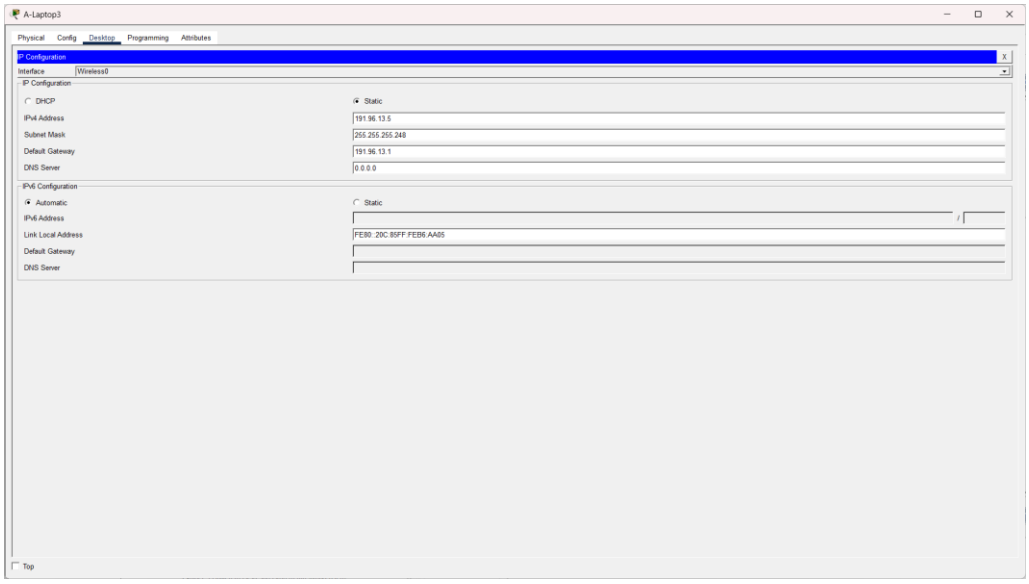

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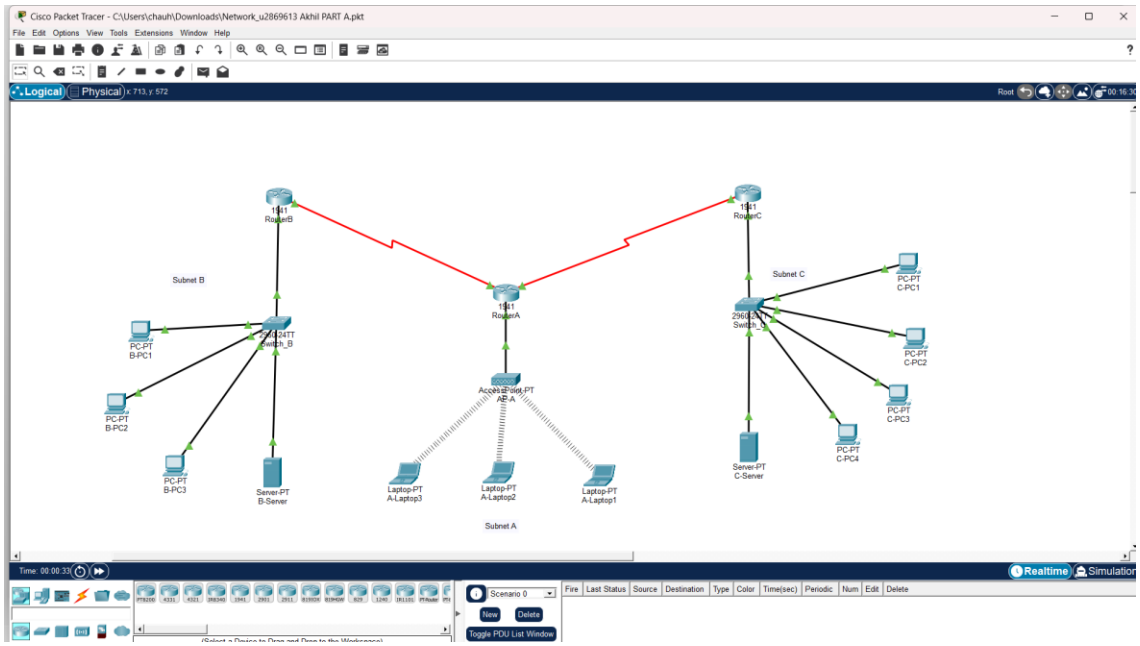

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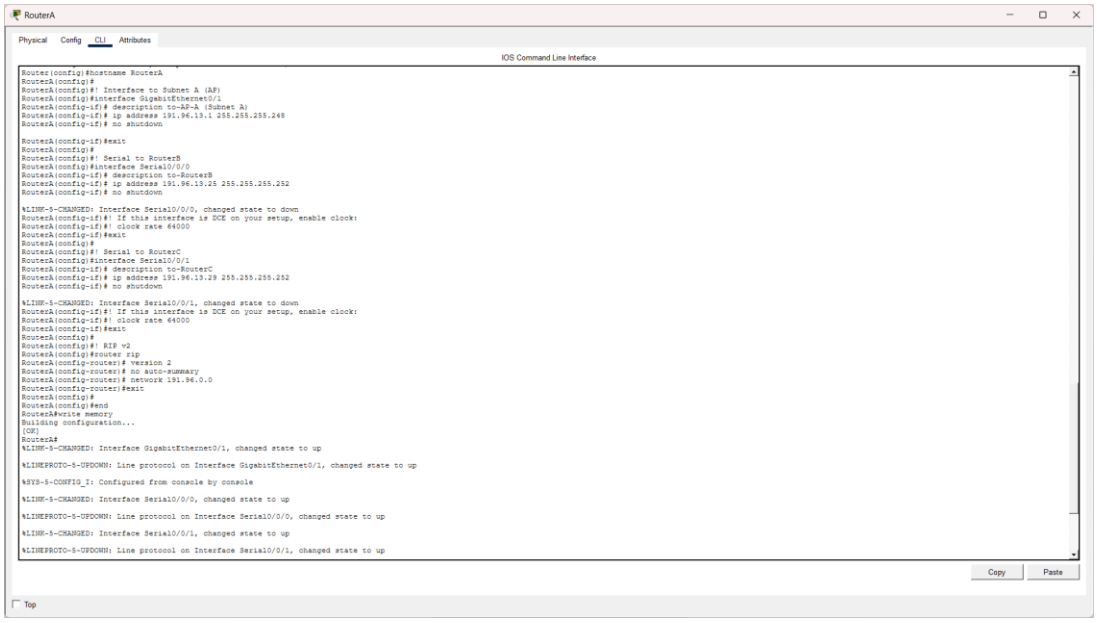


```
Router(config)#hostname RouterA
RouterA(config)#
RouterA(config)#! Interface to Subnet A (AP)
RouterA(config)#interface GigabitEthernet0/1
RouterA(config-if)# description to-AP-A (Subnet A)
RouterA(config-if)# ip address 191.96.13.1 255.255.255.248
RouterA(config-if)# no shutdown

RouterA(config-if)#exit
RouterA(config)#
RouterA(config)#! Serial to RouterB
RouterA(config)#interface Serial0/0/0
RouterA(config-if)# description to-RouterB
RouterA(config-if)# ip address 191.96.13.25 255.255.255.252
RouterA(config-if)# no shutdown

%LINK-5-CHANGED: Interface Serial0/0/0, changed state to down
RouterA(config-if)#! If this interface is DCE on your setup, enable clock:
RouterA(config-if)#! clock rate 64000
RouterA(config-if)#exit
RouterA(config)#
RouterA(config)#! Serial to RouterC
RouterA(config)#interface Serial0/0/1
RouterA(config-if)# description to-RouterC
RouterA(config-if)# ip address 191.96.13.29 255.255.255.252
RouterA(config-if)# no shutdown

%LINK-5-CHANGED: Interface Serial0/0/1, changed state to down
RouterA(config-if)#! If this interface is DCE on your setup, enable clock:
RouterA(config-if)#! clock rate 64000
RouterA(config-if)#exit
RouterA(config)#
RouterA(config)#! RIP v2
RouterA(config)#router rip
RouterA(config-router)# version 2
RouterA(config-router)# no auto-summary
RouterA(config-router)# network 191.96.0.0
RouterA(config-router)#exit
RouterA(config)#
RouterA(config)#end
RouterA#write memory
Building configuration...
[OK]
RouterA#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

%SYS-5-CONFIG_I: Configured from console by console

%LINK-5-CHANGED: Interface Serial0/0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/0/0, changed state to up

%LINK-5-CHANGED: Interface Serial0/0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/0/1, changed state to up
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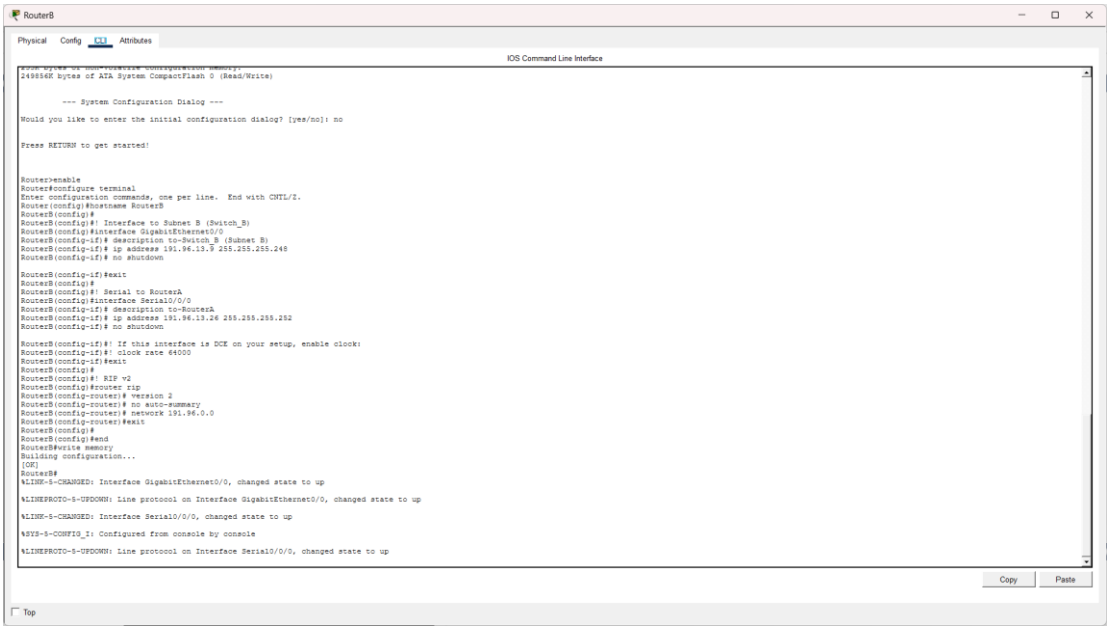

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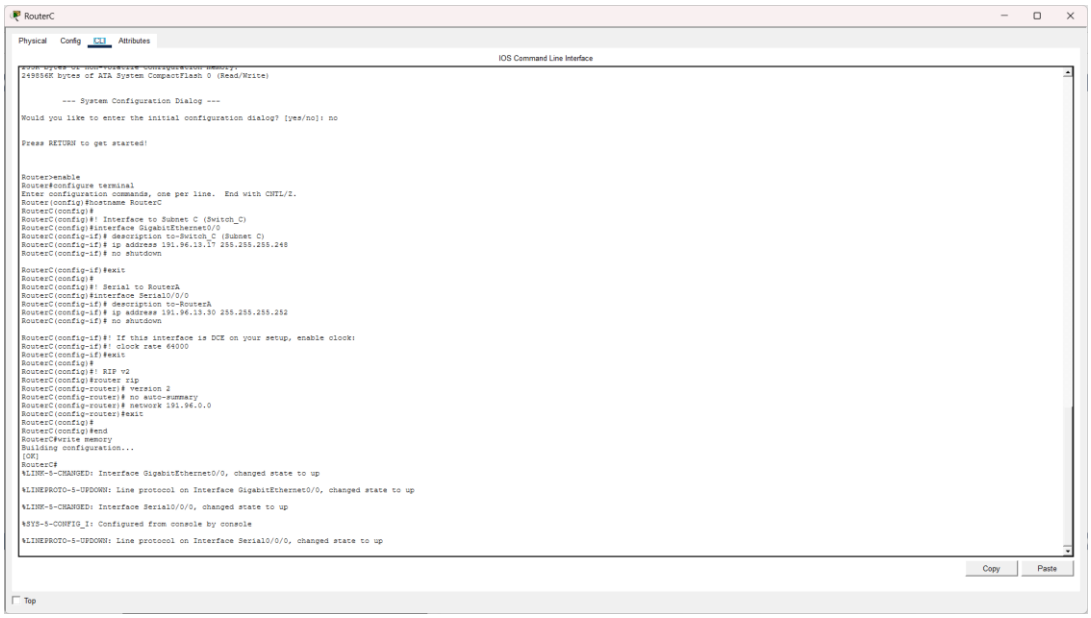

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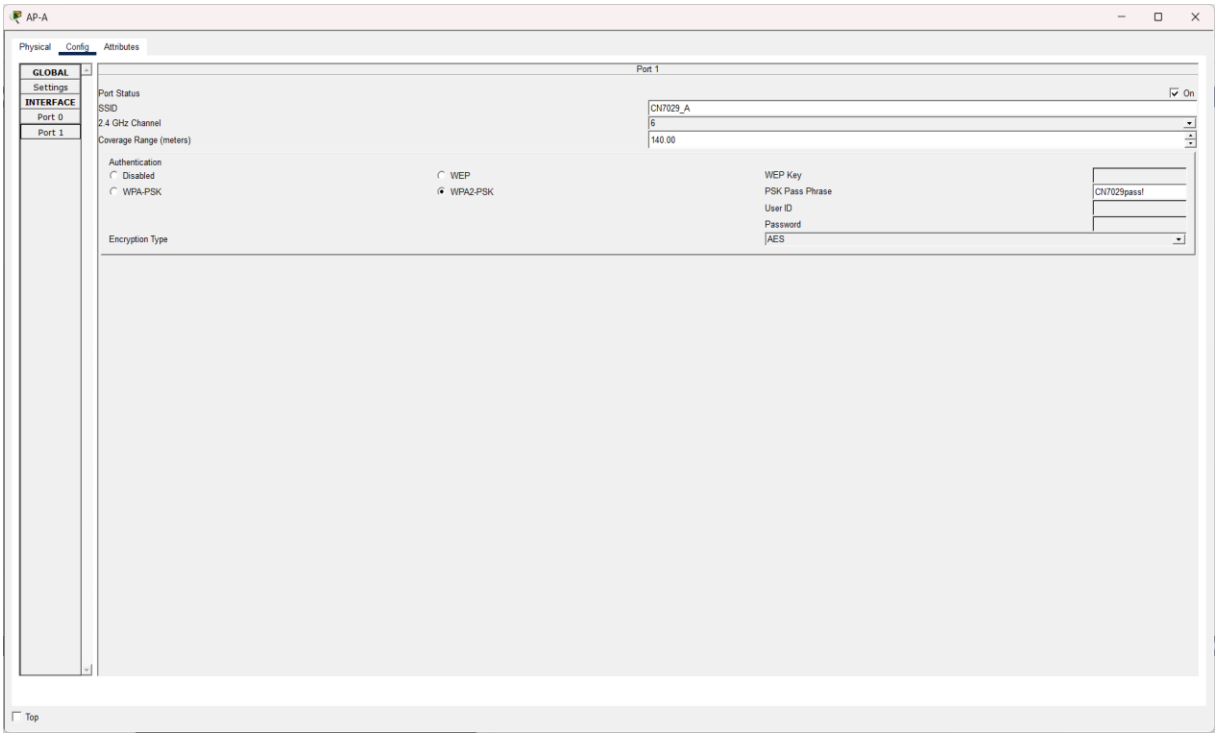
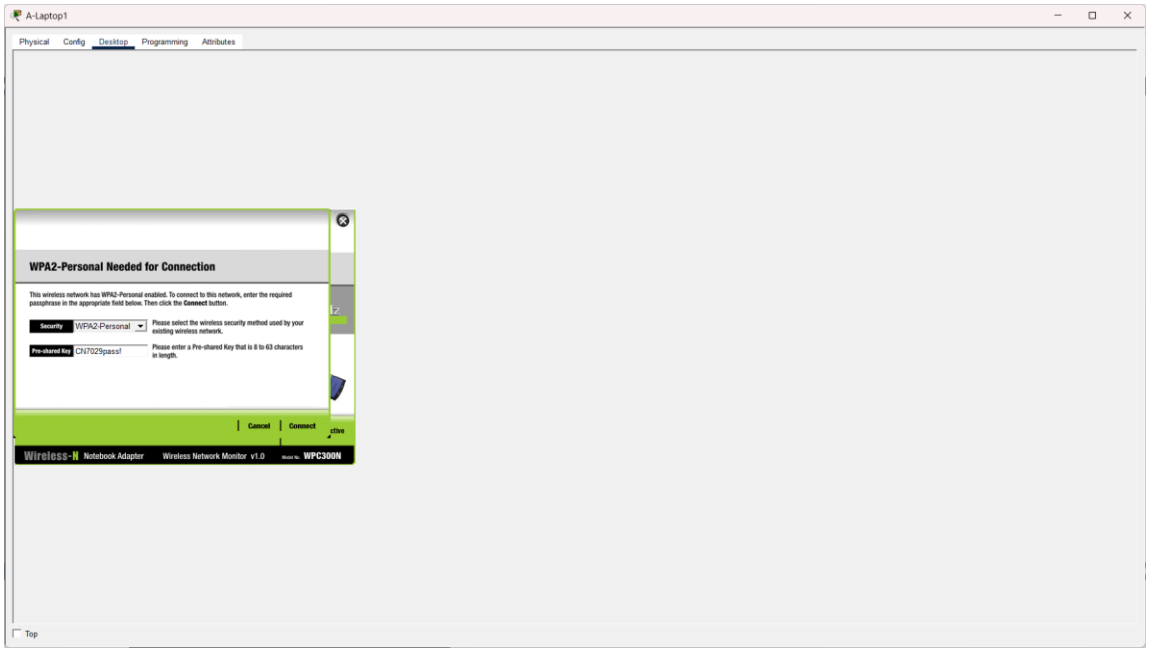

AP-A
Physical
Config
Attributes
GLOBAL
Settings
INTERFACE
Port 0
Port 1
Port 1
Port Status
On
SSID
CN7029_A
2.4 GHz Channel
6
Coverage Range (meters)
140.00
Authentication
Disabled
WPA-PSK
WEP
WPA2-PSK
WEP Key
PSK Pass Phrase
CN7029pass!
User ID
Password
Encryption Type
AES
Top



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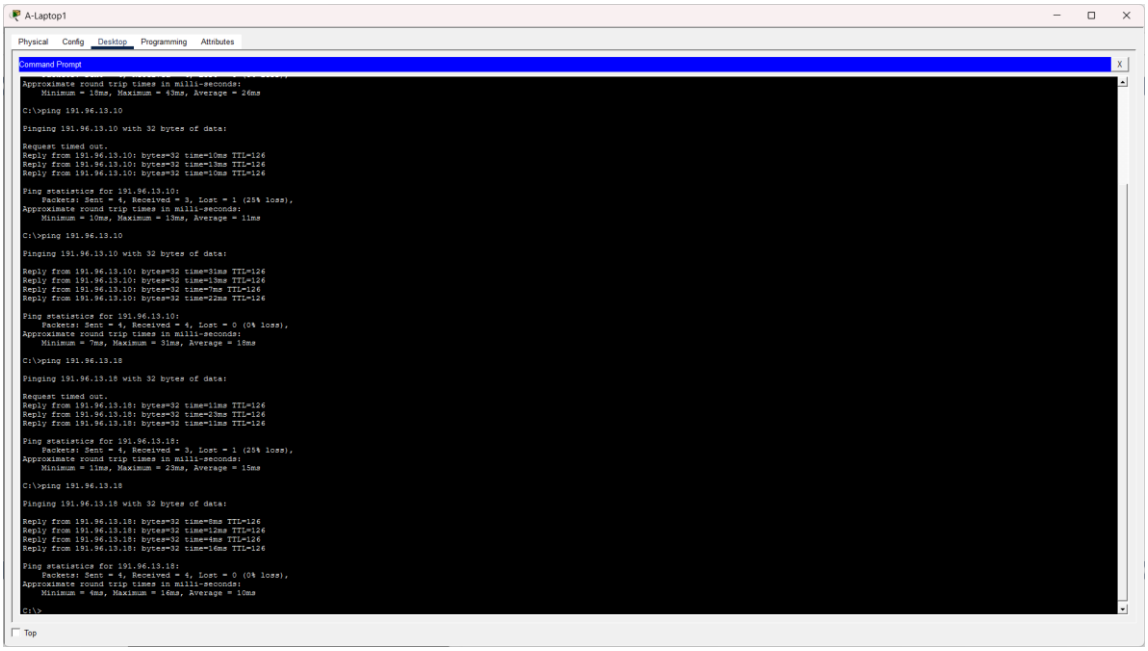

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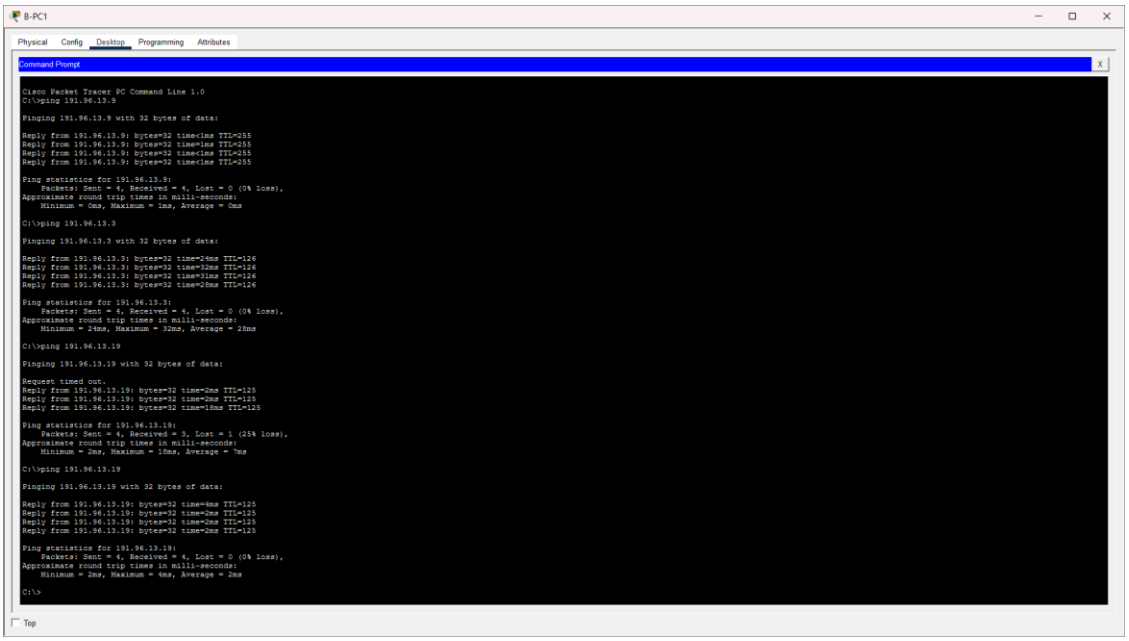

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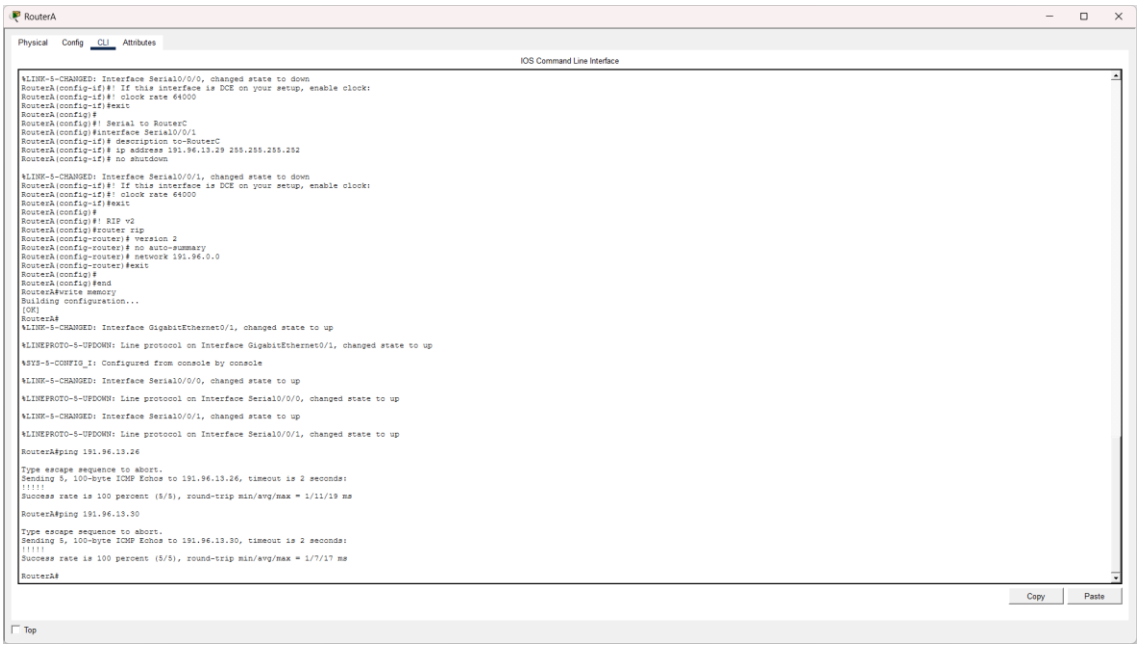

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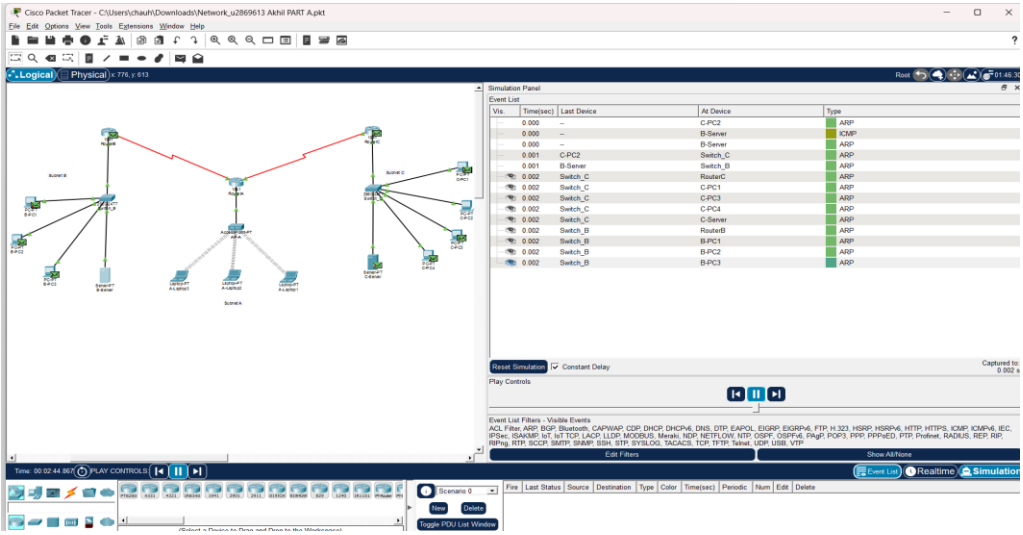

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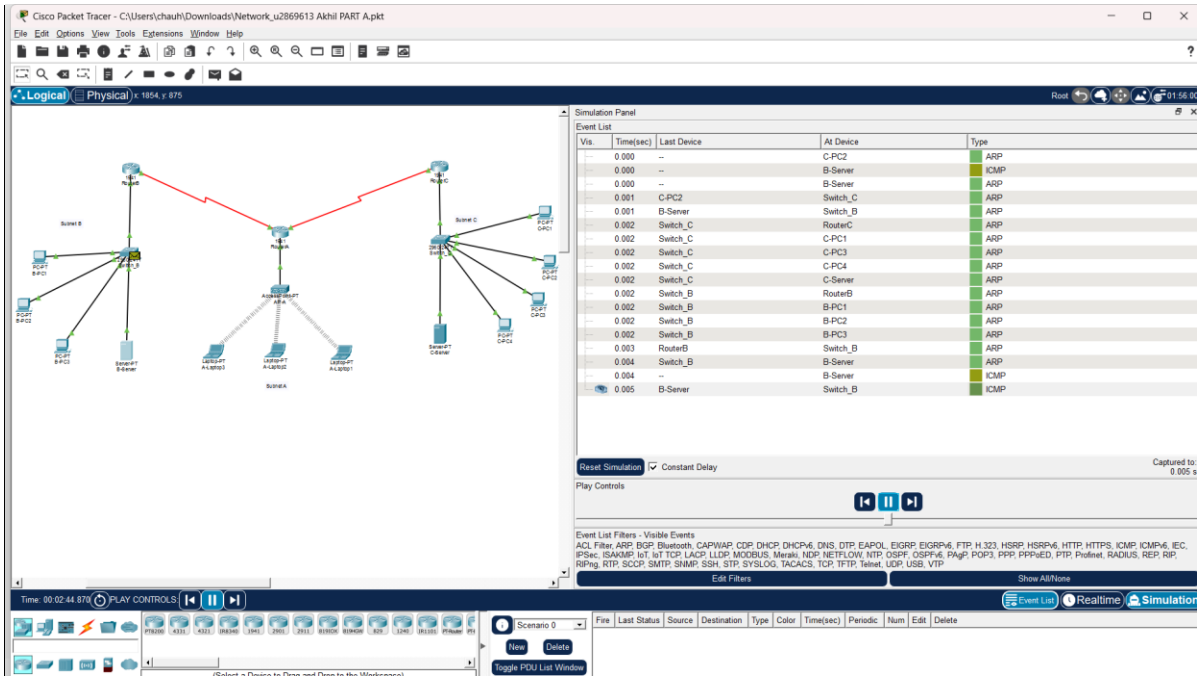

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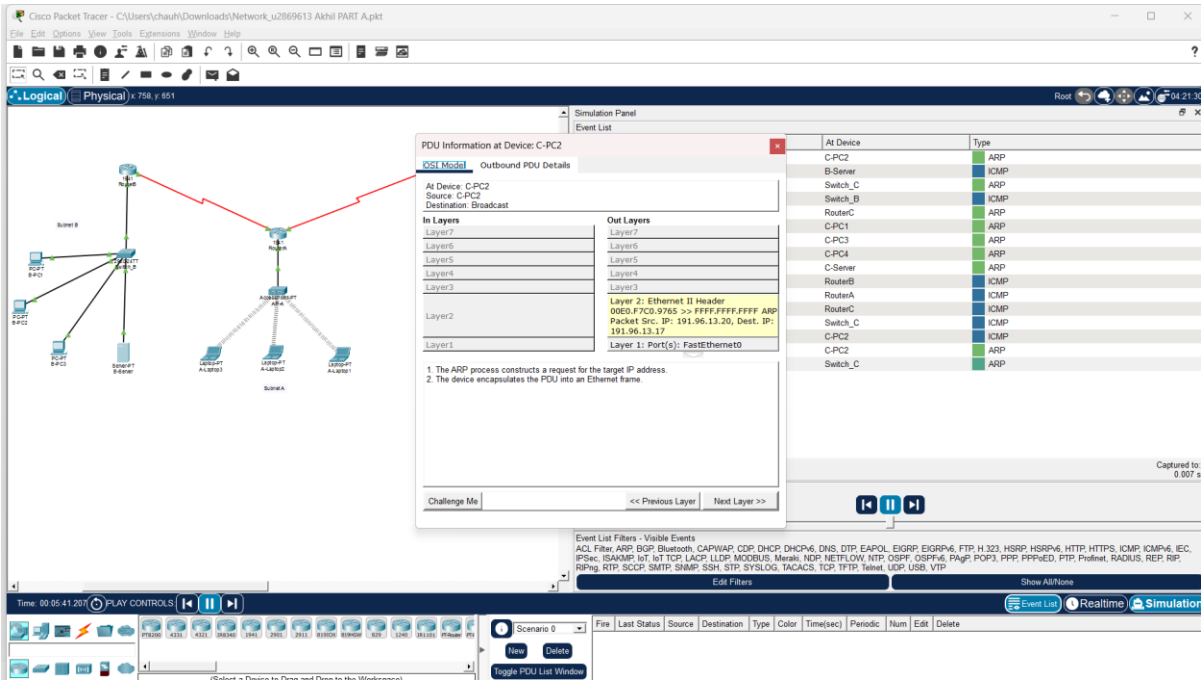

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Subnet	Network Address	Mask	First Host Address	Last Host Address	Broadcast	Bit mask
A (Wireless)	191.96.13.0	255.255.255.248	191.96.13.1	191.96.13.6	191.96.13.7	/29
B (Wired)	191.96.13.8	255.255.255.240	191.96.13.9	191.96.13.22	191.96.13.23	/28
C (Wired)	191.96.13.24	255.255.255.224	191.96.13.25	191.96.13.54	191.96.13.55	/27

[redacted]

Connection Point	Cable / Medium
1. Between Routers and Switches	Copper Straight-Through
2. Between Routers	Serial DCE/DTE
3. Between Router A and Access Point	Copper Straight-Through
4. Between Switch B and Computers	Copper Straight-Through
5. Between Switch C and Computers	Copper Straight-Through
6. Between Switch B and Server B	Copper Straight-Through
7. Between Switch C and Server C	Copper Straight-Through
8. Wireless Devices to Access Point	Wireless Medium

[redacted]

Device	IP Address	Subnet Mask	Gateway
Host B1	**191.96.13.10**	255.255.255.240	**191.96.13.9**
Host B2	**191.96.13.11**	255.255.255.240	**191.96.13.9**
Host B3	**191.96.13.12**	255.255.255.240	**191.96.13.9**
Server B	**191.96.13.13**	255.255.255.240	**191.96.13.9**

[redacted]

Device	IP Address	Mask	Gateway	SSID	Security
Access Point (AP-PT)	**Not Required**	N/A	N/A	**NET-A**	**WPA2-PSK**
Laptop A1	**191.96.13.2**	255.255.255.248	**191.96.13.1**	NET-A	WPA2-PSK
Laptop A2	**191.96.13.3**	255.255.255.248	**191.96.13.1**	NET-A	WPA2-PSK
Laptop A3	**191.96.13.4**	255.255.255.248	**191.96.13.1**	NET-A	WPA2-PSK